Incident Report: Suspicious Login Leading to Account Compromise

Alert Title

Unusual Successful Login from Foreign Location

Summary

A suspicious login was detected for user `hassan.ali` following multiple failed authentication attempts from the same external IP. The login occurred from an unusual geographic location during off-hours using a new device, and no MFA was triggered. This pattern strongly indicates a credential-based attack leading to account compromise.

Initial Findings

- User: `hassan.ali`
- Source IP: `91.203.145.77`
- Time: 02:35 AM
- Event: Successful Login (4624)
- Failed Attempts Before Login: 6
- Location: Russia
- Normal Location: Pakistan
- Device: New (Linux / Chrome)
- MFA: Not triggered

Key Observations

- Login from unusual location (geo anomaly)
- Multiple failed attempts followed by success
- Off-hours login activity
- New device used
- MFA not triggered
- IP associated with data center infrastructure

Investigation

```
let Failed = SecurityEvent
| where EventID == 4625
| where Account == "hassan.ali"
| summarize FailedCount = count(), LastFail = max(TimeGenerated) by
IPAddress;

let Success = SecurityEvent
| where EventID == 4624
| where Account == "hassan.ali"
| project IPAddress, SuccessTime = TimeGenerated;

Failed
| join kind=inner (Success) on IPAddress
| where SuccessTime between (LastFail .. LastFail + 10m)
| project IPAddress, FailedCount, SuccessTime
```

Evidence

- Multiple failed login attempts
- Successful login from same IP
- New device and location
- MFA bypass

Conclusion

This incident is classified as a **Confirmed True Positive (Account Compromise)**.
The attacker successfully gained unauthorized access using credential-based attack techniques.

Severity

High

Response Actions

- User account disabled temporarily

- Password reset enforced
- All active sessions revoked
- Escalated to Tier 2
- Monitoring enabled for suspicious activity

Analyst Note

This incident highlights the importance of enforcing MFA and detecting failed-to-success login patterns. Alerts should be tuned to identify such behavior proactively.